

Talaan E-Invoicing System: Technical Build Plan

Sep 28, 2026 · @Arnold

This plan turns the Talaan prototype into a production system that AAA & Co. can run for SME clients, registered with the BIR and compliant with RR 7-2024, RMC 77-2024 and RMC 98-2026.

1. Purpose and scope

Source of truth. The functional specification is the published prototype (`talaan-demo/index.html`) and the *Talaan E-Invoicing System: User Manual*. Every rule the prototype enforces (computations, checks, approvals, numbering, immutability) is ported to the server, where it becomes binding. The prototype's code can be read as pseudo-code for the business logic; its browser-only parts (storage, sign-in, signing keys) are replaced.

In scope.

- Multi-client service: each client company with its head office and branches.
- Documents: e-invoices (formats B1 to B5), credit memos, Collection Receipts, correction notices, stock documents.
- Discount engine: sale discounts; SC, PWD, NAAC, MOV and solo-parent discounts; provider-defined rules.
- Inventory by branch, customer and item masters, small sales tally, discount register, foreign-currency invoicing and collections.
- Delivery to buyers (email, online view, signed QR), verification service.
- Users, roles, limits, two-factor sign-in, audit and security logs.
- EIS JSON generation and certification; transmission switched on when the BIR requires it.
- Books of accounts (sales and purchase journals, general ledger) needed for CAS registration, which the prototype lacks.

Out of scope for the first release. Payroll, full purchasing and payables, bank reconciliation, and native mobile apps.

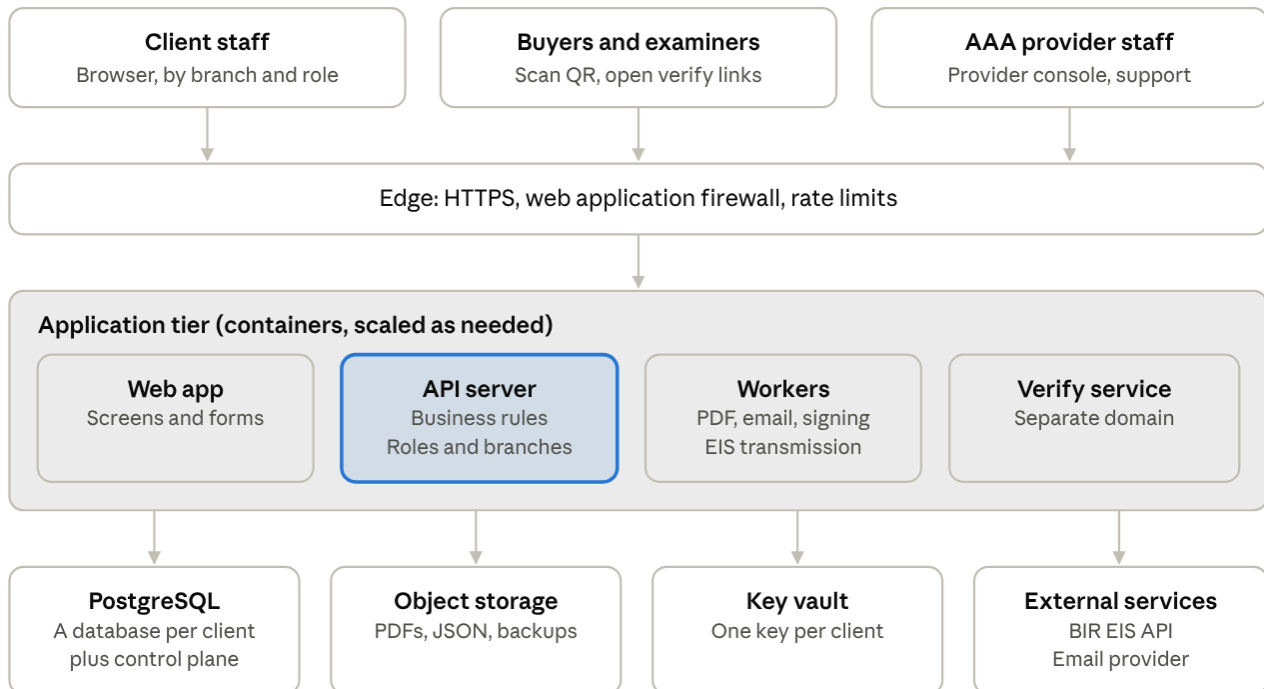
The time constraint. Covered taxpayers must issue e-invoices by December 31, 2026 (RMC 98-2026, IV.1). From today that is about 13 weeks, which is less than the full build in section 8. The plan therefore has a **mandate track**: the smallest registered system that

lets pilot clients issue compliant e-invoices by year-end, followed by the remaining modules.

2. Server architecture

A single multi-client web service run by AAA & Co., with each client company's data in its own database and every business rule enforced on the server.

One service, a separate database per client company



server architecture · 4 layers

Components.

Component	Responsibility
Web app	The screens from the prototype, rebuilt as a single-page app. Holds no business rules of its own; every check is repeated on the server.
API server	All business rules: computations, checks, approvals, numbering, immutability, role and branch access. Stateless, so it can run several copies.

Component	Responsibility
Workers (job queue)	Slow or retryable work: PDF rendering, email delivery, document signing, EIS transmission with retries, nightly backups and reports.
Verify service	The public QR verification page on its own domain (e.g. verify.talaan.ph). Reads only a published summary of each document; no access to client databases.
PostgreSQL	One control-plane database (clients, provider users, statutory rules, design versions, audit of provider actions) and one database per client company (everything in section 4).
Object storage	Issued PDFs, signed JSON, uploaded supporting documents, encrypted backups. Write-once buckets for issued documents.
Key vault	One signing key per client company, never exported; the API asks the vault to sign.

Tenancy. Separate databases per client give the physical separation recommended earlier, simple per-client backup and restore, and a clean exit if a client leaves. Branches are **not** separate databases: every record carries a branch code, and the database's row-level security limits each user to their branches.

Hosting. A managed cloud with a Philippine or nearby Asian region, managed PostgreSQL with point-in-time recovery, and separate production, staging and EIS-certification environments. Where the data sits and who processes it is recorded for the Data Privacy Act.

3. Technology stack

These are recommendations; the IT Division should swap any item for one its team already knows well, since familiarity matters more than the choice itself on this timeline.

Layer	Recommended	Why
Language	TypeScript on both server and browser	The prototype's logic is JavaScript, so the computation engine ports almost line for line and can be shared by server and screens.

Layer	Recommended	Why
API framework	NestJS (Node.js)	Structured modules, guards for roles and branches, good testing support. Alternative: Python with Django if the team is Python-based.
Web app	React with a component library	Matches the prototype's screens; one codebase for desktop and phone browsers.
Database	PostgreSQL 16, managed	Transactions for gapless numbering, row-level security for branches, JSONB for document snapshots, triggers to block edits.
Queue	Redis-backed job queue (e.g. BullMQ)	Retries and scheduling for email, PDF, signing and EIS jobs.
PDF	Headless Chromium rendering the same HTML layouts as the prototype	Keeps printed documents identical to what users preview.
Sign-in	Self-hosted identity server (e.g. Keycloak) or a managed identity service	Password hashing, TOTP and passkeys, lockout, sessions; less security code to write in-house.
Keys and signing	Cloud key management service or hardware security module	Private keys never leave the vault; ES256 signatures for QR codes and JWS for EIS.
Storage	S3-compatible object storage with object lock	Write-once storage for issued documents.
Infrastructure	Containers (Docker), infrastructure as code, CI/CD pipeline	Repeatable environments for production, staging and certification.
Monitoring	Centralized logs, metrics and alerts	Detect failed deliveries, signing errors, queue backlogs and suspicious sign-ins.

4. Database tables

Every issued document is stored with a frozen copy of what was printed, a hash and a signature, and the database itself refuses to change or delete it.

4.1 Control-plane database (provider)

Table	Key columns
clients	id, registered name, TIN, database name, status, modules enabled, onboarding date
provider_users	id, name, role (support, admin), two-factor, active
design_versions	client id, version, logo, colours, font, sizes, BIR reference, effective date, activated by
design_change_requests	id, client id, fields, description, status, BIR reference, history
statutory_rules	code, name, legal basis, effective and expiry dates, rate, VAT treatment, caps, required fields, active
support_access_grants	provider user, client, granted by client, from, to, reason
provider_audit_log	at, provider user, action, detail (append-only)

4.2 Client database: company, branches and people

Table	Key columns
company	registered name, trade name, base TIN, VAT status, PTI number, date, approved system, EIS certification date, sales-reporting flag
branches	code, name, registered address, RDO, BIR notice date, active
series	branch, document type, from, to, next number, authority (PTI or manual booklet)
users	id, username, name, position, role, branch or all, approval limit, discount cap, two-factor, active, last sign-in
customers	id, registered name, TIN with branch code, address, email, tax status (VAT, non-VAT, individual, foreign), country and foreign tax ID for foreign buyers (TIN left empty)

Table	Key columns
<code>items</code>	id, SKU, barcode, description, unit, price, default tax treatment, type, reorder level, SC/PWD coverage, NAAC, MOV, solo-parent, added-rule coverage
<code>item_branch_prices</code>	item, branch, price

`exchange_rates` : currency, date, pesos per unit, source (BAP for US dollars, BSP for other currencies), entered by. One row per currency per day, locked once a document has used it.

4.3 Client database: documents

Table	Key columns
<code>invoices</code>	id, branch, number, format, sales type, nature, date of transaction, issued at, customer, seller snapshot , buyer snapshot , design version, currency, frozen exchange rate (rate, source, date), VAT-inclusive flag, statutory discount (type, ID, name, extra fields, group meal), sale discount, withholding rate, totals, EIS unique ID, references (manual, reissue, additional billing, aggregate), document hash, signature, verification token
<code>invoice_lines</code>	invoice, line no., item, SKU, description, unit, quantity, unit price, line discount, allocated sale discount, tax treatment, statutory choice, statutory benefit, BNPC discount, amount
<code>credit_memo_requests</code>	id, invoice, reason, details, lines (amount and quantity), prepared by, status, history
<code>credit_memos</code>	id, branch, number, invoice, reason, lines, totals, snapshots, prepared by, approved by, timestamps, hash, signature
<code>correction_requests</code> / <code>correction_notices</code>	as for credit memos, with the fields changed (from, to), reason and supporting document

Table	Key columns
collection_receipts	id, branch, number, type (collection or advance), payer snapshot, method, account, currency, amount, collection-date rate, purpose
receipt_allocations	receipt, invoice, amount, applied at, realized forex gain or loss (pesos)
deliveries	document, channel (email, link, QR, printed), recipient, at
small_sales	branch, day, time, description, amount, tax treatment, aggregate invoice
closed_days	branch, day, count, total, aggregate invoices

4.4 Client database: stock, reporting and logs

Table	Key columns
stock_documents	id, type (purchase, return to supplier, adjustment, transfer), branch, to-branch, supplier, supplier document, reason, status, prepared by, approved or received by
stock_document_lines	document, item, quantity, unit cost, quantity received
stock_moves	item, branch, at, kind, quantity, unit cost, source document (written only by the system when a source document is posted)
eis_transmissions	document, attempt, sent at, response, acknowledgment, error
journal_entries / journal_lines	books of accounts: sales and purchase journals and general ledger, generated from documents
security_log	at, user, event, detail, IP address, device (append-only)
access_log	document, viewer, at (examiner and support views)

Accounting export tables: `account_mappings` (posting key, account code, account name), `branch_locations` (branch, location or cost center), `export_batches` (id, user, from, to, summary level, branch, target, format, entry count, total, status, acknowledgment) and `export_entries` (entry ID, batch, period, branch, lines, target reference returned).

4.5 Rules the database enforces

- **No change after issue.** A trigger rejects any UPDATE or DELETE on an issued invoice, credit memo, receipt or notice, and on their lines. Corrections are new rows.
- **Gapless numbering.** The next number is taken inside the same transaction that saves the document, with a row lock on the `series` row, so two cashiers never get the same number and a failed save leaves no gap.
- **Branch security.** Row-level security policies filter every table with a branch column by the signed-in user's branches.
- **Tamper evidence.** Each issued document stores a SHA-256 hash of its frozen content, and the security and audit logs are hash-chained, so a change made outside the application is detectable.
- **Money as integers.** Amounts in centavos, as in the prototype, to avoid rounding drift.
- **Foreign currency.** Amounts are stored in the document's currency, in minor units, with the rate frozen at issue; peso equivalents are computed from that rate and stored with the document. Collections store their own rate and the realized gain or loss per invoice (RMC 12-2024).

5. API outline

A REST API over HTTPS; every call is checked for role, branch and limits on the server, whatever the screen allowed.

Area	Main endpoints	Notes
Sign-in	<code>POST /auth/login</code> , <code>/auth/mfa</code> , <code>/auth/password</code> , <code>/auth/logout</code>	Handled by the identity server; lockout and timeout there
Company and branches	<code>GET/PUT /company</code> , <code>GET/POST /branches</code> , <code>GET/PUT /series</code>	Settings need the administrator role
Users	<code>GET/POST/PUT /users</code> , <code>POST /users/{id}/reset</code> , <code>/unlock</code> , <code>/deactivate</code> , <code>POST /access-reviews</code>	Administrator only

Area	Main endpoints	Notes
Masters	<code>/customers</code> , <code>/items</code> , <code>/items/{id}/branch-prices</code>	Type-ahead search by name, TIN, SKU or barcode
Invoices	<code>POST /invoices/preview</code> (compute and check, nothing saved), <code>POST /invoices</code> (issue), <code>GET /invoices</code> , <code>GET /invoices/{id}</code> , <code>GET /invoices/{id}/pdf</code>	Preview returns format, totals, checks and advisories, exactly as the prototype's editor shows them
Delivery	<code>POST /documents/{id}/deliveries</code> (email, link, printed)	Logged per document
Credit memos	<code>POST /credit-memo-requests</code> , <code>POST .../{id}/approve</code> , <code>.../decline</code> , <code>GET /credit-memos/{id}</code>	Approver limit and "not your own" enforced
Corrections	<code>POST /correction-requests</code> , <code>.../approve</code> , <code>.../decline</code>	Field policy enforced by tax status
Receipts	<code>POST /collection-receipts</code> , <code>GET /collection-receipts</code>	Allocations or advance
Small sales	<code>POST /small-sales</code> , <code>POST /small-sales/close-day</code>	Non-VAT companies only
Stock	<code>POST /stock-documents</code> , <code>.../approve</code> , <code>.../receive</code> , <code>GET /items/{id}/stock-card</code> , <code>GET /reports/inventory-reconciliation</code>	By branch or consolidated
Reports	<code>/reports/discount-register</code> , <code>/reports/branch-totals</code> , <code>/reports/sales-journal</code> , <code>/reports/readiness</code>	CSV and PDF export
Verification (public)	<code>GET verify.talaan.ph/v/{token}</code>	Public summary only; examiner view requires sign-in and is logged
Provider console	<code>/provider/design-requests</code> , <code>/provider/statutory-rules</code> , <code>/provider/support-access</code>	Provider staff with two-factor only
EIS	internal jobs, no public endpoint	JSON build, sign, encrypt, transmit, store acknowledgment

Accounting export: GET/PUT /accounting/mappings, GET /accounting/journal?from&to&level=day|week|month|quarter|year&branch&target (preview), POST /accounting/exports (create a batch as CSV or JSON), POST /accounting/exports/{id}/push (send to a connected NetSuite or SAP Business One), GET /accounting/exports (history).

Foreign currency: GET/POST /exchange-rates (bookkeepers and approvers), GET /reports/fx-summary (the audit summary of foreign-currency transactions) and GET /reports/forex-gains (realized gains and losses). Invoice and receipt endpoints take a currency field and refuse to issue without a rate on file.

6. Security and data privacy

The controls simulated in the prototype become real here, and a few that a browser cannot do are added.

Accounts and access.

- Individual accounts only; passwords stored as salted hashes (Argon2 or bcrypt), never readable.
- Two-factor by authenticator app or passkey, required for approvers, administrators, auditors and all provider staff.
- Lockout after 5 failed attempts, 15-minute idle timeout, server-side sessions that can be revoked.
- Role, branch, approval limit and discount cap checked on every API call; row-level security as a second line in the database.
- Provider support can see a client's data only through a time-limited grant the client approves, and every view is logged.

Data protection.

- TLS for all traffic; encryption at rest for databases, storage and backups.
- One signing key per client in the key vault; keys rotate yearly, and old public keys stay published so older QR codes still verify.
- Daily encrypted backups with point-in-time recovery; a restore test every quarter.
- Records kept for the preservation period the Tax Code and regulations require, including after a client leaves (export handed over, then retained or deleted per contract).

Data Privacy Act.

- AAA & Co. acts as personal information processor for its clients; a data processing agreement with each client.

- A designated data protection officer, a privacy impact assessment, registration with the NPC where required, and a breach response plan that meets the NPC's notification rules.
- The public verification page shows no buyer details; examiner access is signed-in and logged.

Assurance.

- Security review of the code and an external penetration test before go-live and yearly after.
- Alerts on repeated failed sign-ins, denied actions, signing failures and unusual export volumes.
- The ESP rules in RMC 98-2026 (II.5) call for accountable technical, compliance, data-protection and security officers in the Philippines; name them before go-live.

7. EIS integration and certification

The system must pass EIS Certification within 6 months of each client's PTI Electronic Invoice (RMC 98-2026, IV.15), even though live sales reporting waits for the BIR's directive.

1. **Get the official template.** Download the current EIS JSON template (CAS or CRM/POS variant) and technical guide from eis-cert.bir.gov.ph. Replace every placeholder field name used in the prototype and the Python script.
2. **Write the mapping.** A field-by-field table: Talaan field, EIS field, type, length, mandatory, legal basis, validation. Include the foreign-currency fields (currency, rate, source, peso equivalents) and how a foreign buyer without a Philippine TIN is sent (empty TIN, country, foreign tax ID), credit memos, replacement and aggregate invoices, and confirm whether correction notices and Collection Receipts are transmitted.
3. **Build the pipeline** as worker jobs: build JSON from the frozen document, validate against the template, sign (JWS) with the client's key, encrypt as the guide specifies, transmit, store the acknowledgment. Retries with backoff; alerts on rejection.
4. **Certification environment.** A separate deployment wired to the BIR's test endpoints, with test clients, running the 5 mandatory tests (7 with API callback) and keeping evidence of each.
5. **Gate live transmission.** A per-client switch turned on only when the BIR issues sales-reporting guidelines and the client holds a Permit to Transmit (IV.2, IV.15). Until then, JSON is generated, signed and stored for every document so nothing is lost.
6. **The 3-day rule.** Once live, a monitor flags any document not acknowledged within the transmission deadline.

7A. Accounting export to NetSuite, SAP and other software

Talaan posts summarized, balanced revenue-side journal entries to the client's accounting software; purchases, payables and stock adjustments are not exported, and invoice-level detail stays in Talaan as the subsidiary sales journal.

Summary levels. The user picks a date range and a summary level: day, week (Monday to Sunday), month, quarter or year. One entry is produced per period per branch. A period cut by the range is clipped to it and labelled with its actual dates.

What each entry contains (amounts in pesos; foreign-currency documents at their frozen rates):

Source	Debit	Credit
Invoices	Cash (cash sales) or receivables (charge sales); creditable withholding tax withheld by the client (e.g. 10% expanded withholding on professional fees); statutory discounts	Output VAT; sales by class: VATable, zero-rated, VAT-exempt, percentage-tax (net of promotions)
Credit memos	Sales returns and allowances, output VAT	Cash or receivables, creditable withholding tax
Collections	Cash; realized forex loss	Receivables at the invoice rate; realized forex gain
Advances	Cash, then advances from customers when applied	Advances from customers, then receivables
Cost of sales (optional, off by default)	Cost of sales for goods sold; inventory for customer returns	Inventory for goods sold; cost of sales for customer returns (weighted average cost)

Lines are netted per account within the entry. Rounding from currency conversion is absorbed in the largest sales line so every entry balances to the centavo.

Mapping. A mapping table, set once per client, translates each posting key (CASH, AR, CWT, ADV, OVAT, SALES_V, SALES_Z, SALES_E, SALES_P, DISC_ST, RET, FXG, and INV and COGS only when cost of sales is included) to the client's account code and name, and each branch to a location, cost center or profit center. Export is blocked while any key is unmapped or any entry is unbalanced.

Formats.

Target	CSV	JSON / API
Generic	One row per line: entry ID, period, dates, branch, location, account, debit, credit, currency, memo	Entries with lines, totals and source documents
Oracle NetSuite	Journal import columns: External ID, Date, Memo, Account, Debit, Credit, Location, Line Memo	REST record <code>journalEntry</code> (externalId, tranDate, memo, line items with account, debit, credit, location), sent over SuiteTalk REST with OAuth 2.0
SAP Business One	Reference, ReferenceDate, Memo, AccountCode, Debit, Credit, LineMemo, Reference2	Service Layer <code>POST /JournalEntries</code> (ReferenceDate, Memo, Reference, JournalEntryLines)
SAP S/4HANA or ECC (per client)	Generic CSV	Journal Entry – Post API (S/4HANA), or ACC_DOCUMENT IDoc / BAPI_ACC_DOCUMENT_POST (ECC)

Controls.

- **No double posting.** Each entry's ID is fixed: branch, summary level and period dates (e.g. TLN-00000-M-2026-09-01-2026-09-30). It is sent as NetSuite's externalId or SAP's Reference, so the receiving system rejects a repeat.
- **Export history.** Every export is logged with user, range, level, branch, target, entry count and total, and a warning appears when a new export overlaps an earlier one.
- **Period close.** After a month is exported and closed, corrections flow as new documents (credit memos, adjustments) into the next export; exported entries are never regenerated with different amounts.
- **Reconciliation.** A report compares Talaan's sales, VAT, receivables and stock totals with what the accounting system accepted, per period and branch.

Delivery. Download (CSV, JSON), scheduled drop to SFTP, or a direct connector pushing to NetSuite REST or the SAP Business One Service Layer, with the acknowledgment stored against each entry.

7B. Features added to the prototype after the first draft

These are in the current prototype and user guides and are part of the build scope. Each needs the tables and endpoints listed.

Feature	Build notes	Data	Phase
Cashier counter and Cashier role	Counter screen over the same API; role limited to counter permissions; office roles can switch	<code>users.role = CASHIER</code>	P2
Payment details on cash sales	Method (cash, check, e-wallet, bank, card), bank or provider, reference, date; printed under Payment	<code>invoice_payments</code>	P2
Charge sales: unpaid notice, terms, due date	Terms per client (due on receipt to Net 60, or a date); due date = transaction date + terms; overdue status	<code>customers.terms</code> , <code>invoices.terms</code> , <code>invoices.due_date</code>	P2
Receivables aging	Buckets: not due, 1-30, 31-60, 61-90, over 90 days; as-of date; CSV	view over invoices, credit memos, collections	P3
Manual withholding	Preset rate, any rate or fixed amount; client default rate; cashier only for named clients with a reason	<code>invoices.wht_mode</code> , <code>wht_rate</code> , <code>wht_amount</code> , <code>wht_reason</code> ; <code>customers.wht_rate</code>	P2
VAT class per line and reclassification	Class printed per line; change needs reason and document, cashier needs approver (password), logged	<code>invoice_lines.tax_change</code> (from, reason, document, by, approved_by)	P2

Feature	Build notes	Data	Phase
SC/PWD split per line	Group-meal proration or per-line "for the SC" flag; exemption tied to the ID	<code>invoice_lines.for_beneficiary</code>	P4
Buyer email on the invoice; automatic email on issue	Company setting plus per-client opt-out; logged per delivery	<code>customers.auto_email</code>	P2
Client information form (portal)	Public form with 2303 upload and privacy consent; office approval creates the customer	<code>portal_submissions</code> , object storage for uploads	P3
Cashier invoice history and totals	Own invoices only, by date and search, with totals	query over <code>invoices.issued_by</code>	P2
Item categories, filters, pages; CSV import	Server-side search and paging; CSV template, validation report, import of valid rows	<code>items.category</code> , <code>import_batches</code>	P3
Storage indicator	Usage per client against plan; alerts at 80/90/100%; never blocks invoicing	<code>storage_usage</code> , <code>plans</code>	P7
Philippine Standard Time	All dates computed in Asia/Manila regardless of device	server time zone setting	P1

Feature	Build notes	Data	Phase
Help assistant and support chat	Help articles with an assistant; support threads with context; provider inbox	support_threads , support_messages , help_articles	P7

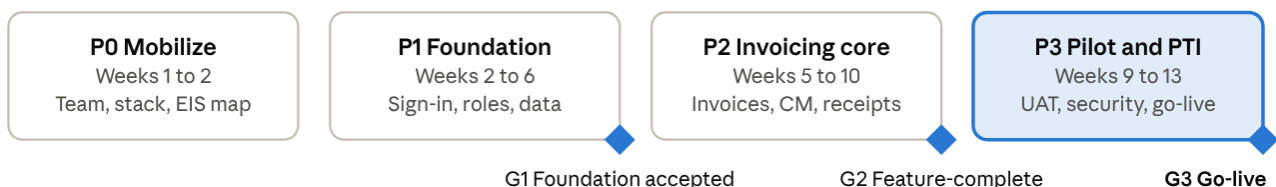
Updated EIS script. `talaan_eis_transmit.py` (version 2) reads Talaan's structured data, checks each document (TIN formats, VAT arithmetic, withholding, statutory discount details, foreign-currency peso equivalents, unpaid flag on charge sales, VAT class changes with reasons), maps fields through `FIELD_MAP`, signs (ES256 JWS), encrypts (JWE) and writes or transmits, with a CSV log. Its field names and envelope stay placeholders until mapped to the official template.

8. Sequence of work

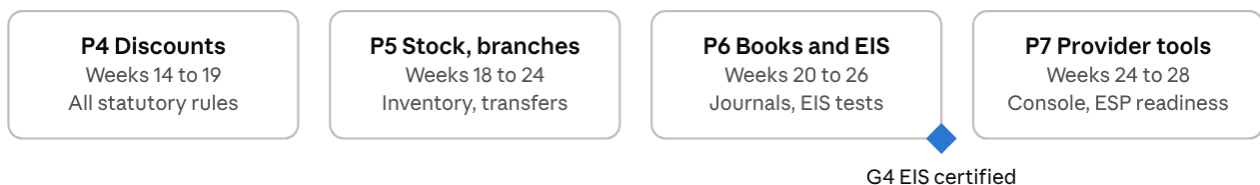
Build what pilot clients need to issue compliant e-invoices first, file the PTI applications early, and add the remaining modules after go-live.

Invoicing goes live by December 31; the rest follows by April 2027

Mandate track: September 28 to December 31, 2026



After go-live: January to April 2027



sequence of work · 8 phases, 4 gates

Mandate track (weeks 1 to 13).

- P0 Mobilize.** Staff the team, fix the stack, set up the three environments and CI/CD. Download the EIS template and draft the field mapping. Confirm with the RDO/LT Office what the PTI and system registration require, and choose the 2 to 3 pilot clients.

2. **P1 Foundation.** Client databases, identity with two-factor, roles, branches and series, company settings, customers and items, security log, the no-change triggers and gapless numbering. **Gate G1:** the IT lead and you accept the foundation, and the go/no-go on the December 31 date is taken.
3. **P2 Invoicing core.** Port the computation engine from the prototype: VAT boxes, formats B1 to B5, sale and SC/PWD discounts, the ₱1,000 rule, withholding. Then issuance, PDF, email and QR delivery, signing and the verification page, downtime replacement, credit memos with approval, Collection Receipts. **Gate G2:** every reference case in section 9 passes.
4. **P3 Pilot and PTI.** Security review and penetration test, user acceptance with the pilot clients, training with the User Manual, follow-up on the PTI Electronic Invoice applications filed in week 7, go-live. **Gate G3:** pilot clients issuing live by December 31.

After go-live (weeks 14 to 28).

5. **P4 Discounts.** NAAC, MOV, solo-parent and provider-defined rules, discount register, small sales tally, correction notices, foreign-currency invoicing with exchange rates, collections and the realized forex report (moved into P2 if a pilot client sells in foreign currency).
6. **P5 Stock and branches.** Inventory by branch, transfers with receipt confirmation, branch prices, branch and consolidated reports.
7. **P6 Books and EIS.** Sales and purchase journals and general ledger for CAS; the accounting export (section 7A) with downloads first and NetSuite and SAP Business One connectors next; the EIS certification tests. **Gate G4:** EIS certification, due within 6 months of each client's PTI.
8. **P7 Provider tools.** Provider console, design-change workflow, support access grants, ESP officers and documents once the BIR's ESP rules are issued.

Critical path. The BIR evaluates a PTI application within 20 working days of complete documents (IV.12). Counting back from December 31, applications must be filed by mid-November, allowing for the December holidays, which means the system documentation and a working build must exist by about week 7, while P2 is still under way. If G1 shows that date cannot hold, decide then on a fallback for the pilot clients rather than in December.

9. Team, testing and acceptance

Team for the mandate track.

Role	People	Focus
Technical lead and architect	1	Architecture, database, reviews, EIS mapping
Back-end developers	2 to 3	API, computation engine, workers, signing
Front-end developer	1 to 2	Screens, PDF layouts
QA engineer	1	Automated tests from the reference cases, regression
DevOps and security	0.5 to 1	Environments, backups, key vault, monitoring
Tax subject-matter expert	part-time	Rules, sign-off on computations and formats (AAA & Co. tax team)
Project manager	part-time	Gates, pilot clients, BIR filings

Reference cases. Each was computed and checked in the prototype; the server must reproduce the same figures to the centavo before G2.

No.	Scenario	Expected result
1	B1 invoice: 120 reams at ₱215, 6 toners at ₱2,850 less ₱500, 4 zero-rated crates at ₱1,500, 1% withholding	VATable ₱42,400.00, VAT ₱5,088.00, zero-rated ₱6,000.00, withholding ₱484.00, due ₱53,004.00
2	Senior citizen, reading glasses ₱1,680 VAT-inclusive	VAT-exempt ₱1,500.00, discount ₱300.00, due ₱1,200.00
3	10% sale discount on ₱7,850 of VATable goods	VATable ₱7,065.00, VAT ₱847.80, due ₱7,912.80
4	Senior citizen during a 10% sale: paracetamol, rice, paper, toner	Paracetamol gets 20% SC; rice and paper keep the sale discount; due ₱6,725.00
5	Three rice purchases in one week under one SC ID	5% BNPC of ₱62.50, ₱62.50, then ₱0.00 (₱125 cap)
6	Group meal ₱4,000 VAT-inclusive, 4 diners, 1 senior	Discount ₱178.57, due ₱3,714.29
7	Athlete: basketball ₱1,800 plus paper ₱215	Discount ₱360.00, VAT ₱241.80 on full price, due ₱1,896.80

No.	Scenario	Expected result
8	Solo parent: formula milk, diapers, toner during a 10% sale	Discount ₱258.00 VAT-exempt; toner keeps the sale discount; due ₱5,194.80
9	Medal of Valor: paracetamol ₱350	Discount ₱70.00, due ₱322.00
10	Credit memo on case 1: return 10 reams and 2 toners	Total credited ₱8,528.50, stock back in
11	Transfer of 4 toners to Makati, 3 received	Makati stock +3, shortage of 1 flagged
12	Approver with ₱50,000 limit on an ₱80,000 credit memo	Refused; a higher approver can approve

Foreign-currency cases. (13) USD 1,200 zero-rated invoice at ₱58.10: peso equivalent of amount due ₱69,720.00. (14) USD 3,000 invoice booked at ₱57.85 and collected at ₱58.10: realized forex gain ₱750.00. (15) An invoice dated a day with no rate on file is refused. (16) A foreign buyer saved without a country is refused; with country and address it needs no TIN, prints "None (foreign buyer, country)", and a zero-rated services line shows the Sec. 108(B)(2) reminder.

Beyond the reference cases. Tests that issued documents cannot be changed through the API or directly in the database; that numbering has no gaps or duplicates under many simultaneous cashiers; that branch users cannot read other branches; that a tampered PDF fails verification; and load tests at peak-day volumes of the pilot clients.

Acceptance. Each gate is signed off by the technical lead and by the tax expert. Go-live also needs written user acceptance from each pilot client and a clean penetration-test report.

10. Open decisions and risks

These need a decision from you before or at Gate G1, because each changes the scope or the date.

Item	Why it matters	Recommendation
Registration route	A full CAS needs books of accounts (P6); an invoicing system may be registered sooner under RMC 98-2026 III.4	Register first as an e-invoicing system; add the CAS books in P6
PTI timing	20 working days for evaluation; December holidays	File by mid-November; agree a fallback for pilot clients at G1

Item	Why it matters	Recommendation
EIS template	Field names in the prototype are placeholders	Map to the official template in P0; certification tests in P6
ESP status	RMC 98-2026 promises separate ESP rules	Build to ESP-style controls now; appoint the officers in II.5
Correction notices	Not yet a BIR-named document	Keep the conservative policy (address, non-VAT and B2C only) until the sales-adjustment rules come out
Medal of Valor tax treatment	No BIR regulation found; RA 9049 mentions a tax credit for admission fees	Keep as "Checked"; confirm before relying on it for deductions
Advance payments	Q31 covers collection after the sale, not advances	Keep Collection Receipt first, invoice on delivery; review when adjustment rules issue
Hosting and data location	Data Privacy Act, client comfort	Philippine or nearby Asian region, recorded in each data processing agreement
Team availability	13 weeks is tight	Staff the full team in week 1, or move non-pilot clients to 2027

Foreign currency. No invoicing rule found expressly requires peso equivalents on a foreign-currency invoice; printing them is a conservative choice so VAT in pesos is visible. Decide whether to keep it, and whether foreign currency belongs in P2 for any pilot client that exports.

Biggest risk. The December 31 date depends on BIR processing that AAA & Co. does not control. Keep the mandate track small, file early, and treat every feature not needed by the pilot clients on day one as a 2027 item.